CS 594 Project Progress Report — Govind Rao
Solar Forecasting for Dynamic Pricing and Energy Allocation Using Weakly Coupled MDP

1 Dataset and Preprocessing

We use SKIPP'D, containing 64×64 RGB sky images and one-minute photovoltaic (PV) power measurements collected at Stanford University during 2017–2019 [4]. The benchmark contains 363,375 observations in HDF5 format with separate timestamp files. We processed the earliest 60 available days.

A Python script using NumPy, pandas, and h5py checks array alignment, sorts observations chronologically, and excludes invalid measurements and ambiguous duplicate timestamps. No invalid PV/timestamp records or duplicate timestamps were found in the source, and no invalid pixel values were found in the selected images.

Each example uses 16 consecutive images and power readings from $t - 15$ through t to predict power at $t + 15$ minutes. Examples are sampled every five minutes within uninterrupted recording segments; windows crossing gaps or day boundaries are excluded. New chronological splits assign approximately 70%, 15%, and 15% of days to training, validation, and testing.

The loader scales pixels to $[0, 1]$, standardizes power using training-target statistics only, and encodes time of day using sine and cosine features. Sequence indices, targets, split assignments, and normalization statistics are saved in `prepared_skippd.npz`. Images are loaded on demand from the original HDF5 file.

Table 1: Results for the processed 60-day subset.

Quantity	**Count**
Selected observations	47,069
Training examples	4,365
Validation examples	853
Test examples	1,028
Total forecasting examples	6,246

2 Simulation Environment

A simulator has been implemented for three microgrids with price-responsive demand and local batteries. At each one-minute step, a policy selects prices and electricity allocations under a shared supply constraint. Demand follows an assumed price-response model with independent local random shocks. Battery updates account for consumption, charging/discharging limits, and efficiency losses.

Solar supply can be replayed from SKIPP'D, converting power to energy for each step, alongside an assumed fixed electricity contract. Rewards include revenue, procurement costs, battery wear, and unmet-demand penalties. Fixed-price, myopic, and exploratory baselines observe current supply without future outcomes. Parameters are illustrative; synthetic checks are complete, while replay on the downloaded data remains to be verified.

3 Use in the Project

The image sequences will train a CNN to forecast solar supply. Recorded PV measurements will provide realized supply trajectories for policy evaluation, while simulated customer responses supply the demand information absent from SKIPP'D.

Forecasts will inform a WCMDP policy that plans pricing and battery use across future steps. Comparisons against the implemented baselines will test whether improved forecasts increase profit and

reduce unmet demand. Remaining work includes visual inspection, CNN training, verification of data replay, and dynamic policy integration.

4 Optimization Formulation and Candidate Methods

Let $s_{i,t}$ describe microgrid i's battery and demand conditions, and let z_t contain common supply information and forecasts. Policy π selects prices $p_{i,t}$ and energy allocations $q_{i,t}$ to maximize expected cumulative profit:

$$\begin{aligned} \max_{\pi} \quad & \mathbb{E}_{\pi}\left[\sum_{t=1}^{T}\sum_{i=1}^{M} r_i(s_{i,t}, z_t, p_{i,t}, q_{i,t})\right] \\ \text{subject to} \quad & \sum_{i=1}^{M} q_{i,t} \le C_t, \qquad \forall t, \\ & (p_{i,t}, q_{i,t}) \in \mathcal{A}_i(s_{i,t}, z_t), \qquad \forall i, t. \end{aligned}$$

Here, C_t is available energy, $\mathcal{A}_i$ specifies feasible local actions, and r_i is expected revenue minus costs and unmet-demand penalties. Local transitions are conditionally independent given common supply information and actions. We will discretize states and actions and investigate three approaches.

Lagrangian relaxation. Following Adelman and Mersereau [1], multipliers on the shared supply constraint allow decomposition into local dynamic programs. The resulting value approximations and bounds will guide feasible joint decisions.

Feasibility network relaxation. Nadarajah and Cire [2] represent linking constraints through a network preserving feasible combinations of component actions within an LP relaxation. Its self-adapting construction adjusts network size to the constraint structure. We will investigate its application to discretized electricity allocations.

Delayed allocation. Nadarajah and Cire [3] identify feasible joint actions before allocating probability flows, using convex hulls of feasible action subsets in marginalized flow models. This supports policies satisfying linking constraints by construction and reduces reliance on subsequent feasibility corrections.

Comparisons will consider profit, unmet demand, computation time, and, where available, gaps between policy values and upper bounds. Github repository is available at [5]

References

[1] Daniel Adelman and Adam J. Mersereau. Relaxations of weakly coupled stochastic dynamic programs. *Operations Research*, 56(3):712–727, 2008.

[2] Selvaprabu Nadarajah and Andre A. Cire. Self-adapting network relaxations for weakly coupled Markov decision processes. *Management Science*, 71(2):1779–1802, 2025.

[3] Selvaprabu Nadarajah and Andre Augusto Cire. Delayed allocation in marginalized flow models for weakly coupled Markov decision processes, 2026. Working paper.

[4] Yuhao Nie, Xiatong Li, Andea Scott, Yuchi Sun, Vignesh Venugopal, and Adam Brandt. Skipp'd: A sky images and photovoltaic power generation dataset for short-term solar forecasting. *Solar Energy*, 255:171–179, 2023.

[5] Govind Rao. CS 594 Project: Solar forecasting for dynamic pricing and energy allocation, 2026. Source code and experiment outputs.